

Forecasting Stock Market Variance Using Machine Learning

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1 Empirical Specification

Quarter t realized stock market variance is $MV_t = \sum_i^{N_t} r_{t,i}^2$, where $r_{t,i}$ is the daily return in day i and N_t is the number of daily returns in quarter t . We try to forecast MV_t or $\log(MV_t)$ using machine learning and various features. One specification is a generalized Midas model. The expected quarter t variance is a function of (squared) daily returns in the past 252 days: $\mathbb{E}(MV_t) = f(r_{t,-1}^2, \dots, r_{t,-252}^2)$. We can also include the Goyal and Welch predictor variables as well as those considered in Schwert (1989) and the JFE paper in the forecast. We use the out-of-sample R^2 to evaluate the model performance.